WHERE ARBITRARY-SCALE SUPER-RESOLUTION GAINS COME FROM: HIGH-FREQUENCY CAPACITY AND SCALE-PAIRED LEARNING

Anonymous authors

Anonymous institution

ABSTRACT

Arbitrary-scale super-resolution (ASISR) networks decode encoder features at continuous coordinates and condition the decoder on the target pixel size, or cell. We ask where their gains come from. In frequency normalised by the low-resolution (LR) Nyquist limit, u, recovery profiles are shared across four decoder designs, and capacity—longer training, a stronger encoder or a wider decoder MLP—improves mainly bands near and above the Nyquist limit ($u \gtrsim 0.75$), also under a Gaussian-blur degradation. The cell is often thought to lose its effect at large scales. This impression is an artefact of fixed cell multipliers, which shrink the actual decoder-input step as $2(m-1)/s$: at equal steps, the cell-induced output change at $\times 6$ matches or exceeds that at $\times 2$ in 11 of 12 conditions. The cell also matters during learning: native cell outperforms dimension-preserving constant-cell and shuffled-cell training by 0.100 and 0.097 dB, and both controls are equivalent to removing the cell within ± 0.03 dB. ASISR gains thus concentrate where learned priors fill bands near and above the Nyquist limit, the cell keeps acting at inference, and correctly paired scale information improves learning. Predictions were registered before each experiment; failures are reported.

***Index Terms*—** Arbitrary-scale super-resolution, implicit representation, frequency recovery, scale conditioning

1. INTRODUCTION

Continuous-coordinate super-resolution decodes local image features at arbitrary output locations. LIIF [1] conditions this decoder on target pixel size, or *cell*; subsequent designs use local Fourier estimators [2], integrated positional encoding [3], implicit attention [4], neural operators [5], Gaussian splatting [6] or heat-field filtering [7]. Which parts of the recovered signal benefit from these choices, and what information does cell contribute?

Prior work links performance to model size and training [8] and studies spectral decoding's efficiency–accuracy trade-off [9]. HyRA analyses nonlinear compensation of low-pass phase distortion [10]; LIIF, integrated encoding and heat fields connect cell to edges and filtering [1, 3, 7]. We locate capacity gains and distinguish cell-induced output change, error cost and training benefit.

We argue that ASISR gains come mainly from learned capacity in bands near and above the LR Nyquist limit, that the cell keeps acting at inference at all tested scales, and that correctly paired scale information provides a training benefit. Three findings support this.

1. **Capacity gains concentrate near and above Nyquist.** Tested decoder forms share recovery profiles (spread ≤ 0.019), while capacity gains concentrate from $u \approx 0.75$ upward; longer training changes lower bands by at most 0.0013.
2. **Cell responses do not fade with scale.** At equal decoder-input steps, the output change is sustained or larger at $\times 6$ than at $\times 2$ in 11/12 conditions; declining fixed-multiplier costs reflect shrinking steps and error alignment.
3. **Correct scale pairing supplies the training gain.** Native cell beats constant and shuffled cell by 0.100 and 0.097 dB, and both controls are equivalent to no cell.

Table 2 lists every registered prediction, including failures.

2. SHARED RECOVERY PROFILES AND CAPACITY

Models and measurement. LIIF, LTE and SC-INR (an LTE-derived experimental variant with analytic sinc cell weights) use the EDSR-baseline architecture [11], separately learned encoder weights and a common DIV2K [12] recipe: scales from $\mathcal{U}(1, 4)$, 1000 epochs and three seeds. GaussianSR uses an external 2000-epoch EDSR checkpoint and provides an additional comparison. Matched 200-epoch LIIF arms change the encoder to RDN [13] or SwinIR [14], reduce the MLP from 4×256 to 2×64, or remove cell. These comparisons match recipes; parameter counts and compute differ.

Let $r = \max(|f_x|, |f_y|)$ be frequency in cycles per output pixel and $u = 2sr$ its value relative to the input Nyquist limit at scale s. Recovery in band b is $R_b = 1 - V_b/E_b$, where error energy V_b and target energy E_b are pooled over images before taking the ratio. Original crop analyses use DIV2K 0869–0900, 240^2 centre crops, quarter-u bins, anti-aliased bicubic inputs and a four-pixel shave. Full-image recovery additionally covers Set14 [15], BSD100 [16] and Urban100 [17]. We evaluate $\times 2/3/4/6$; $\times 6$ lies outside the training scale range.

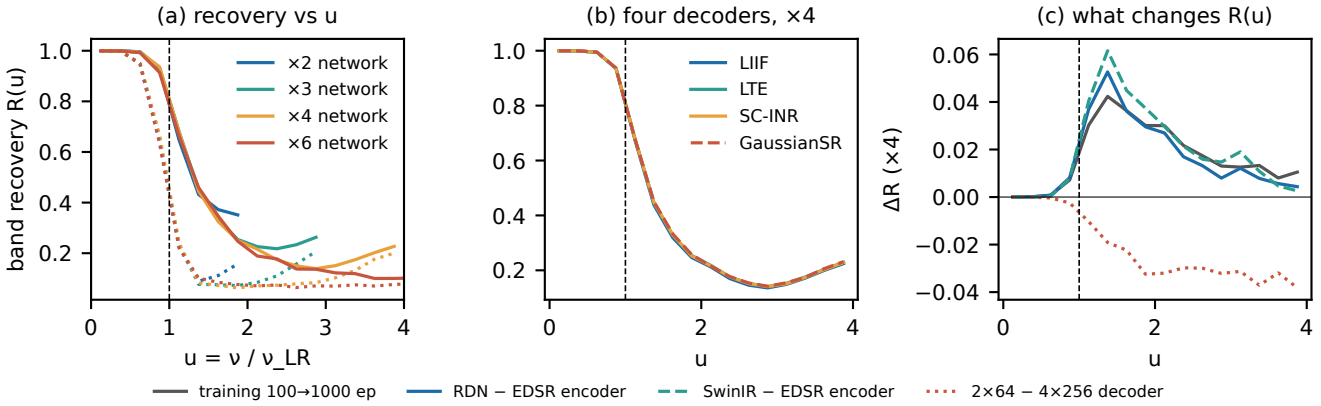


Fig. 1. Shared recovery profiles and frequency-localised capacity effects. (a) Three decoder families and three seeds at four scales; dotted curves are bicubic. (b) Decoder comparison at $\times 4$; GaussianSR is an external checkpoint with a different recipe. (c) Effects of training duration, encoder and MLP capacity. The plotted points represent frequency bands.

A shared recovery regime. Across $\times 2/3/4$, the three matched decoder families have a maximum scale-specific band spread of 0.010, attained at $\times 4$, in the registered $0.5 \leq u < 2$ comparison (paired image-bootstrap 95% interval $[0.007, 0.016]$ at that scale). With GaussianSR included, all 54 full-image dataset–scale–band comparisons stay below 0.019. Closely shared profiles extend across the tested decoder forms and image collections (Fig. 1a,b), although the stricter cross-scale-collapse prediction fails its registered tolerance gates.

Capacity improves the incompletely recovered bands. Longer training, stronger encoders and a larger MLP change recovery primarily from $u \approx 0.75$ upward (Fig. 1c). From epoch 100 to 1000, gains in $1 \leq u < 2$ range from 0.0245 to 0.0464 across the three decoder families and $\times 2/3/4$, while changes below $u = 0.75$ are at most 0.0013. Matched RDN and smaller-MLP experiments replicate the high-frequency improvement and loss, respectively, with a second training seed. Decoder form and decoder capacity therefore have distinct empirical effects.

After retraining with Gaussian blur ($\sigma = 0.6s$ HR pixels) and non-antialiased bicubic resizing, RDN improves PSNR at all scales but changes recovery below $u = 0.75$ by at most 0.0006, failing the registered lower-band-gain prediction. This one-seed result extends low-frequency saturation to a second degradation. Table 1 and Fig. 3 show benchmark and visual effects.

3. CELL RESPONSES PERSIST AS SCALE INCREASES

Equal multipliers imply unequal interventions. For output side length L, the implementations multiply the supplied cell $2/L$ by the LR feature-grid size L/s. The decoder receives $c = 2/s$ on each axis. Multiplying cell by m therefore gives the per-axis step

$$h = \frac{2(m-1)}{s}, \qquad m = 1 + \frac{hs}{2}. \tag{1}$$

For fixed $m = 2$, this step falls from 1 to $1/3$ between $\times 2$ and $\times 6$. A declining perturbation cost consequently combines changing input step with changing network response.

For native prediction P, perturbed prediction P_h, target T, native error $e = T - P$ and output change $\delta = P_h - P$, the MSE change is

$$D = \|e - \delta\|^2 - \|e\|^2 = \underbrace{\|\delta\|^2}_{J} - 2\,\mathrm{Re}\langle e, \delta\rangle. \tag{2}$$

We report J, D and the alignment term in RGB-mean MSE units. Small cost can arise from an output change that corrects existing error. The dB cost additionally depends on native MSE V through $10\log_{10}(1 + D/V)$.

Sustained equal-step response. At the same decoder-input step, cell-induced output changes generally do not weaken with scale and are often larger (Fig. 2). Across four datasets and three steps, $J_6/J_2 \geq 1$ in 11/12 conditions; point estimates span 0.94–2.47. The individual 95% interval lower bounds span 0.81–1.57 and upper bounds 1.08–4.48; these ranges summarise 12 intervals, not a pooled interval. The smallest estimate is Urban100 at $h = 0.25$, 0.94 $[0.81, 1.08]$, and the largest is Set14 at $h = -0.25$, 2.47 $[1.47, 4.48]$. This endpoint pattern is descriptive; the registered tests are in Table 2.

The frozen experiment measures all 246 full images from DIV2K 0869–0900, Set14, BSD100 and Urban100 with LIIF/LTE/SC-INR, two seeds and $h \in \{-0.25, 0.25, 0.5\}$, using paired within-dataset image bootstrap intervals. Matched image support, preprocessing, 10,000 resamples and multiplicity rules are documented in the repository; Table 2

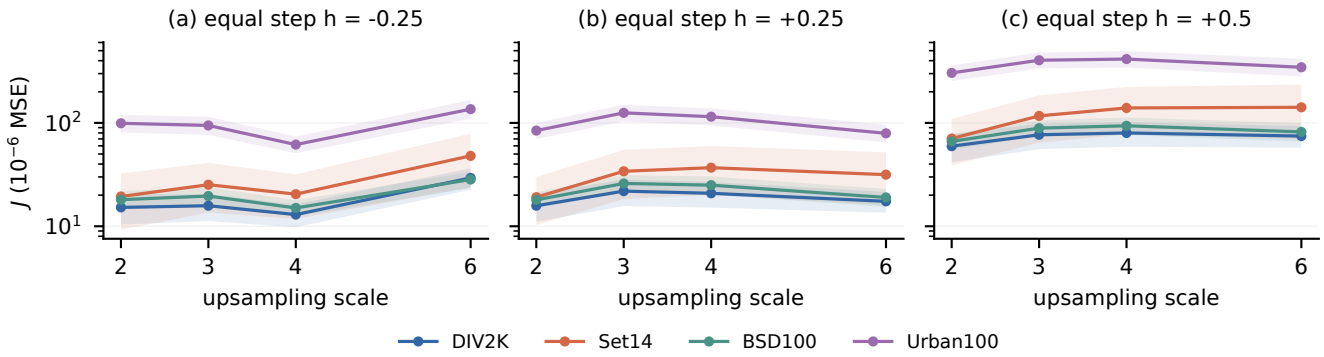


Fig. 2. Equal-step output responses need not decrease with scale. Each curve averages three decoder families and two seeds on one full-image dataset. Shading shows pointwise 95% paired-image bootstrap intervals, conditional on these checkpoints; the logarithmic axis reports J. Endpoint ratios are descriptive; the frozen prediction gates are reported separately in Table 2.

Table 1. Mean full-image Y-PSNR change (dB) relative to native EDSR-LIIF: four datasets, $\times 2/3/4/6$, matched 200-epoch arms and two seeds. Shuffled cell uses native cell at evaluation. †One seed. Parameter counts and compute differ for capacity changes.

Change	$\times 2$	$\times 3$	$\times 4$	$\times 6$
MLP 2×64	-0.25	-0.26	-0.28	-0.24
no cell input	-0.13	-0.09	-0.11	-0.10
constant cell	-0.13	-0.08	-0.09	-0.10
shuffled cell	-0.12	-0.09	-0.09	-0.09
RDN encoder	$+0.35$	$+0.29$	$+0.28$	$+0.25$
SwinIR encoder†	$+0.46$	$+0.42$	$+0.42$	$+0.34$

separately reports the frozen existence, condition-wise and Set14 endpoint tests, including the failed Set14 prediction.

Response and cost can move in opposite directions. On BSD100 at $h = 0.5$, $J_6/J_2 = 1.24$ $[1.06, 1.44]$, but D falls from 79.42 to 40.77 in 10^{-6} MSE (paired change -38.65 $[-55.13, -22.01]$; descriptive 95% intervals, without multiplicity correction). Across datasets at this step, $r \geq 0.375$ accounts for 43.5–50.7% of J at $\times 2$, but only 2.4–3.0% at $\times 6$; the middle band $r \in [0.1875, 0.375)$ accounts for 60.5–66.7% at $\times 6$.

These finite interventions compare responses across changing features and native conditions. Negative steps leave the training cell interval at some scales, and native $\times 6$ is outside it; marginal support is reported for each condition.

Scope of a scalar explanation. A retrospective real-gain fit gives $Q = \sum_b (1 - w_b)^2 R_b E_b$, fitting w_b to native and perturbed outputs. In the original DIV2K bands, doubled-cell Q/D is 0.44, 0.72, 0.81 and 2.50 at $\times 2/3/4/6$. Recovery and fitted-filter changes jointly account for 41.4% of the $\times 2 \to \times 6$ MSE-cost decline (Set14: 41.5%); alignment and non-scalar terms complete the accounting. Projections, image rankings and negative cases remain in the repository.

Table 2. Registered predictions and outcomes: satisfying checks and overall pass (P) or fail (F). Stage 17 uses Bonferroni-corrected one-sided bounds; its 11/12 endpoint count in the text is descriptive. Training counts are per decoder (12 high-band bins, 9 low-band bins). *Best cell within 10% of native (20/27), smaller cell costs more (33/144), $\geq 80\%$ of cost at $u \geq 1$ (311/414), external cell locus (2/4; 3/4).

Prediction	Outcome
decoder spread ≤ 0.03 (crops; full)	36/36; 54/54 P
training gain at $1 \leq u < 2$; $\lvert\Delta R\rvert \leq .01$ at $u < .75$	12/12; 9/9 P
strict $R(u)$ collapse (crops; full)	45/54; 53/72 F
blur moves gains below $u = 0.75$	0/3 F
small MLP harmless at large scales	F
external RDN-BTC; EDSR gain	0/12; 0/4 F
fixed-multiplier cost declines; peak r	9/9; 36/36 P
$Q/D \in [.7, 1.3]$ at $\times 3/4$	6/6 P
four further fixed-multiplier predictions*	F
no-cell loss decays to negligible cost	F
17: $\geq 1/36$ corrected increases $> 10\%$	11/36 P
17: increase in all 12 conditions	12/12 P
17: both positive-step Set14 endpoints	1/2 F
18: native advantage (constant; shuffle)	P; P
18: no-cell equivalence (constant; shuffle)	P; P

4. WHAT CELL CONTRIBUTES DURING LEARNING

Correct pairing supplies the training gain. Native cell exceeds constant-cell and shuffled-cell training by 0.100 dB $[0.061, 0.149]$ and 0.097 dB $[0.058, 0.145]$, respectively (97.5% paired-image intervals). Both seeds favour native cell in each comparison. Both controls also pass the registered no-cell equivalence tests: no-cell minus constant is -0.005 dB $[-0.014, 0.003]$, and no-cell minus shuffle is -0.009 dB $[-0.017, -0.001]$ (95% intervals), wholly within ± 0.03 dB. Thus all four predictions pass (Table 2). Retaining the extra inputs without correct scale pairing does not recover the native-cell benefit.

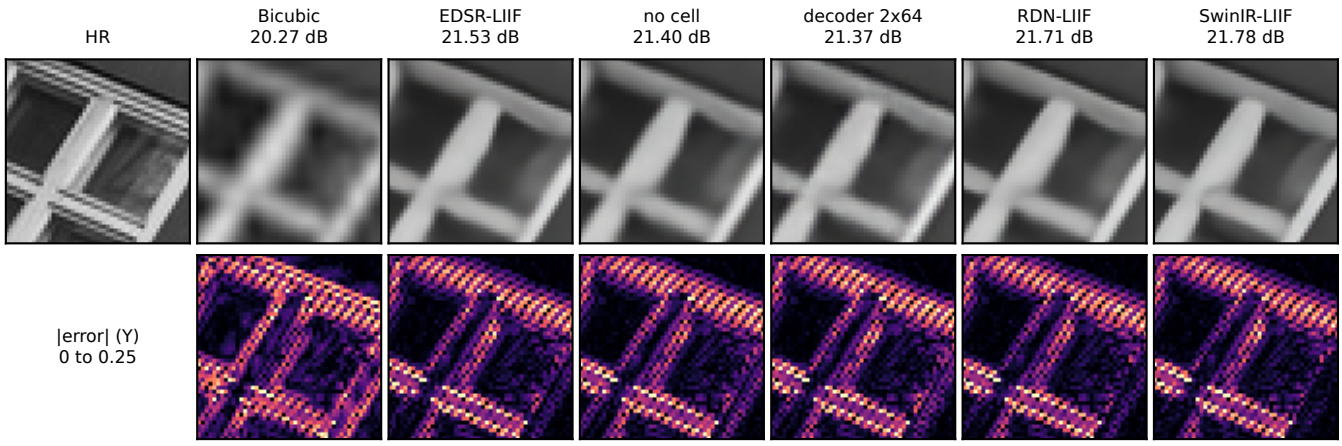


Fig. 3. Rule-selected $\times 4$ recovery example, Urban100 img_036. Before visual inspection, the image is selected at the upper-median rank of whole-image RDN–EDSR Y-PSNR gain (51st of 100); the 48×48 window maximises HR gradient energy on a stride-eight grid. Top: reconstructions and window Y-PSNR. Bottom: absolute Y error, shared range $[0, 0.25]$. The fixed example illustrates the existing capacity and no-cell arms.

Dimension-preserving controls. Both new EDSR-LIIF arms retain the native decoder's two cell inputs and parameter count, under the same 200-epoch recipe and two seeds. The constant arm receives the training mean actual cell coordinate, $c_0 = 0.9242188453$, at every query and evaluation geometry; its input weights can be absorbed into the first-layer bias. The shuffled arm receives another sample's complete cell vector through a batch derangement, preserving the empirical marginal while removing correct pairing. Separate random generators preserve data and initialisation streams; assignment audits verify the interventions.

Frozen scoring uses epoch 200 regardless of validation and reevaluates native/no-cell baselines identically: full-image Y-PSNR on Set5/Set14/BSD100/Urban100 at four scales, plus DIV2K crop $R(u)$. Predictions require native-minus-control ≥ 0.05 dB in the dataset- and scale-balanced mean, positive effects in both seeds and corrected intervals above zero; equivalence requires the entire 95% interval inside ± 0.03 dB. Image bootstrap intervals condition on the two seeds; full protocols and independent verification of all 50,328 records are in the repository.

Frequency and inference-policy checks. The recovery difference $R_{\text{native}} - R_{\text{control}}$ is small below $u = 0.75$: its absolute quarter-band difference stays below 0.00048 for both controls at all four scales. At $\times 2$, mean differences over $1 \leq u < 2$ are 0.0146 $[0.0068, 0.0245]$ for constant and 0.0165 $[0.0100, 0.0246]$ for shuffle (descriptive 95% intervals). Higher-scale region intervals often include zero, so these crop results do not imply a uniform bandwise advantage. For the shuffled model, switching among native, mean and four independent marginal-cell inference policies changes balanced benchmark PSNR by only 0.00014 dB across all six policies. Its loss is therefore not recovered by these test-time cell assignments.

These controls support correctly paired explicit scale information as the source of the benefit in this recipe. They do not identify a unique internal mechanism: queries and features retain other scale cues, and shuffling also injects training noise. All original training runs completed; an authorised evaluation-only recovery resolved a CUDA-initialisation failure using unchanged weights and frozen evaluation/scoring code. The original failure and recovery provenance are retained.

5. IMPLICATIONS AND SCOPE

For model design, learned capacity paid off near and above the LR Nyquist limit, and the tested decoder forms changed that range little: a new decoder or cell design should be judged by whether it moves $R(u)$ there. For analysis, cell probes should hold the decoder-input step fixed and report response and error alignment separately, because fixed-multiplier costs conflate both. For training, the cell is worth keeping even at large scales, but only when it is correctly paired with the target scale.

The evidence covers the tested architectures, recipes and two degradations. New measurements were specified prospectively on previously studied images and checkpoints; intervals condition on the trained seeds, and the ablations do not identify a unique internal mechanism.

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